

cardona-qft-methods: table evidence

Tables

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																				
cardona-qft-methods_ TAB_001	43	■ 1.000	$\begin{tabular}{t l l l l l l l l l } \hline \backslash(d\backslash) & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \hline \backslash(\epsilon\backslash) & 1 & -1 & 1 & 1 & 1 & -1 & 1 & 1 \\ \backslash(\epsilon'\backslash) & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ \backslash(\epsilon''\backslash) & 1 & & -1 & & 1 & & -1 & \end{tabular}$	<table><tr><td>d</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>ϵ</td><td>1</td><td>-1</td><td>1</td><td>1</td><td>1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>ϵ'</td><td>1</td><td>1</td><td>-1</td><td>-1</td><td>-1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>ϵ''</td><td>1</td><td></td><td>-1</td><td></td><td>1</td><td></td><td>-1</td><td></td></tr></table>	d	0	1	2	3	4	5	6	7	ϵ	1	-1	1	1	1	-1	1	1	ϵ'	1	1	-1	-1	-1	-1	1	1	ϵ''	1		-1		1		-1		$[\pi(a), \pi(b)^\circ] = 0, \quad [[\mathcal{D}, \pi(a)], \pi(b)^\circ] = 0, \quad \forall a, b \in \mathcal{A}$ $\pi(a)^\circ = J\pi(a^*)J^{-1}$ is a representation of the opposite alge $\mathcal{A}$-summable (or has metric dimension d) if the singular val ike $\mu_n(\mathcal{D}^{-1}) = \mathcal{O}(n^{-1/d})$. regular if $\mathcal{A}$ and $[\mathcal{D}, \mathcal{A}]$ are in the domain of δ^n for all $n \in$
d	0	1	2	3	4	5	6	7																																	
ϵ	1	-1	1	1	1	-1	1	1																																	
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ϵ''	1		-1		1		-1																																		
cardona-qft-methods_ TAB_002	181	■ 0.546	$\begin{array}{l} \begin{array}{l} \text{\textbf{L}}_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^ae^b}{3l^2})e^c \\ \text{\textbf{E}}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^ae^b}{l^2})T^c \\ \text{\textbf{P}}_4^{Lor} = R^a{}_b R^b{}_a \\ \text{\textbf{N}}_4 = T^a T_a - e^a e^b R_{ab} \\ \text{\textbf{P}}_4^{U(1)} = FF \\ \text{\textbf{P}}_4^{SU(4)} = Tr[\textbf{A}d\textbf{A} + \frac{2}{3}\textbf{A}\textbf{A}\textbf{A}] \end{array} \\ \text{\textbf{L}}_3^{Lor} = \omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a \\ \text{\textbf{L}}_3^{Tor} = e^a T_a \\ \text{\textbf{L}}_3^{U(1)} = AdA \\ \text{\textbf{L}}_3^{SU(N)} = Tr[\textbf{A}d\textbf{A} + \frac{2}{3}\textbf{A}\textbf{A}\textbf{A}] \end{array}$	(not rendered)	topological invariants, <table><tr><th>$D = 3$ CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td>$L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^ae^b}{3l^2})e^c$</td><td>$E_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^ae^b}{l^2})T^c$</td><td>$SO(4),$ $SO(3,1)$ or $SO(2,2)$</td></tr><tr><td>$L_3^{Lor} = \omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a$</td><td>$P_4^{Lor} = R^a{}_b R^b{}_a$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{Tor} = e^a T_a$</td><td>$N_4 = T^a T_a - e^a e^b R_{ab}$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{U(1)} = AdA$</td><td>$P_4^{U(1)} = FF$</td><td>$U(1)$</td></tr><tr><td>$L_3^{SU(N)} = Tr[\textbf{A}d\textbf{A} + \frac{2}{3}\textbf{A}\textbf{A}\textbf{A}]$</td><td>$P_4^{SU(4)} = Tr[\textbf{F}\textbf{F}]$</td><td>$SU(N)$</td></tr></table> Table 1: Three-dimensional gravitational CS Lagrangians, their related	$D = 3$ CS Lagrangians	Characteristic classes	Groups	$L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^ae^b}{3l^2})e^c$	$E_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^ae^b}{l^2})T^c$	$SO(4),$ $SO(3,1)$ or $SO(2,2)$	$L_3^{Lor} = \omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a$	$P_4^{Lor} = R^a{}_b R^b{}_a$	$SO(2,1)$	$L_3^{Tor} = e^a T_a$	$N_4 = T^a T_a - e^a e^b R_{ab}$	$SO(2,1)$	$L_3^{U(1)} = AdA$	$P_4^{U(1)} = FF$	$U(1)$	$L_3^{SU(N)} = Tr[\textbf{A}d\textbf{A} + \frac{2}{3}\textbf{A}\textbf{A}\textbf{A}]$	$P_4^{SU(4)} = Tr[\textbf{F}\textbf{F}]$	$SU(N)$																		
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$L_3^{Lor} = \omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a$	$P_4^{Lor} = R^a{}_b R^b{}_a$	$SO(2,1)$																																							
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Identifier	Page	Conf.	LaTeX source	Rendered	Image																																				
cardona-qft-methods_ TAB_003	182	0.562	$\begin{tabular}{t}{ l l } \hline \text{\(D=7 \mathrm{CS}\)} \\ \text{\(Lagrangians \& Characteristic classes\)} \\ \hline \text{\(L_{\text{Lor}}=\omega(d\omega)^3+\frac{8}{3}\omega^3(d\omega)^2+\cdots+\frac{1}{5}\omega^5\)} \\ \text{\(d\omega^3+\cdots+\frac{4}{7}\omega^4\)} \\ \text{\(R^{\{a\}}_{\{b\}}R^{\{c\}}_{\{d\}}R^{\{d\}}_{\{a\}}\)} \\ \text{\(L^I_{\text{Lor}}=\left(\omega^a_{\{a\}}\omega^b_{\{b\}}d\omega^c_{\{c\}}\right)\omega^d_{\{d\}}\)} \\ \text{\(R^{\{a\}}_{\{b\}}R^{\{c\}}_{\{d\}}R^{\{d\}}_{\{a\}}\)} \\ \text{\(L^I_{\text{Lor}}=\left(e^aT_a\right)\left(R^a_bR^b_a\right)\)} \\ \text{\(R^{\{a\}}_{\{b\}}R^{\{c\}}_{\{d\}}R^{\{d\}}_{\{a\}}\)} \\ \hline \end{tabular}$	(not rendered)	two Lorentz-Pontryagin, and one torsional CS forms, <table><tr><th>$D = 7$ CS Lagrangians</th><th>Characteristic classes</th></tr><tr><td>$L_7^{Lor} = \omega(d\omega)^3 + \frac{8}{3}\omega^3(d\omega)^2 + \cdots + \frac{1}{5}\omega^5$</td><td>$\mathbf{P}_8^{Lor} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$</td></tr><tr><td>$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$</td><td>$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$</td></tr><tr><td>$L_4^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$</td><td>$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$</td></tr></table> Table 9: Seven dimensional gravitational CS Lagrangians, their related	$D = 7$ CS Lagrangians	Characteristic classes	$L_7^{Lor} = \omega(d\omega)^3 + \frac{8}{3}\omega^3(d\omega)^2 + \cdots + \frac{1}{5}\omega^5$	$\mathbf{P}_8^{Lor} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$	$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$	$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$	$L_4^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$	$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$																												
$D = 7$ CS Lagrangians	Characteristic classes																																								
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$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$	$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$																																								
$L_4^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$	$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$																																								
cardona-qft-methods_ TAB_004	205	0.993	$\begin{tabular}{t}{ l l l l l l l l } \hline n \& \theta \& 1 \\ \& 2 \& 3 \& 4 \& 5 \& 6 \& 7 \\ \hline \text{\(\varepsilon \)} \& 1 \& -1 \& -1 \& -1 \& 1 \& 1 \\ \text{\(\varepsilon' \)} \& 1 \& -1 \& 1 \& 1 \& 1 \& -1 \& 1 \& 1 \\ \text{\(\varepsilon'' \)} \& 1 \& \& -1 \& \& 1 \& \& -1 \& \& \\ \hline \end{tabular}$	<table><tr><td>n</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>ε</td><td>1</td><td>1</td><td>-1</td><td>-1</td><td>-1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>ε'</td><td>1</td><td>-1</td><td>1</td><td>1</td><td>1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>ε''</td><td>1</td><td></td><td>-1</td><td></td><td>1</td><td></td><td>-1</td><td></td></tr></table>	n	0	1	2	3	4	5	6	7	ε	1	1	-1	-1	-1	-1	1	1	ε'	1	-1	1	1	1	-1	1	1	ε''	1		-1		1		-1		$[[D, a], b^0] = 0 \quad \forall a, b \in \mathcal{A}.$ table of signs listed above is modeled on the case of ordinary where it reflects the properties of the mod eight periodicity, gy and real K-theory. However, imposing that real struct
n	0	1	2	3	4	5	6	7																																	
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cardona-qft-methods_ TAB_005	281	0.574	$\begin{tabular}{t}{ l l } \hline \text{\(L\)} \& \text{\(p p p p\)} \\ \hline k \text{\(k k\)} \& \text{\(\bar{L} \)} \\ \hline \end{tabular}$	<table><tr><td>L</td><td>$pppp$</td></tr><tr><td>$k k k k$</td><td>L</td></tr></table>	L	$pppp$	$k k k k$	L	gives the subal, ora, expressed i <table><tr><td>L</td><td>P</td><td>Q</td></tr><tr><td>K</td><td>L</td><td>S</td></tr><tr><td>S</td><td>Q</td><td>R</td></tr></table>	L	P	Q	K	L	S	S	Q	R																							
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cardona-qft-methods_ TAB_006	281	0.887	$\begin{tabular}{t}{ l l l } \hline \text{\(L\)} \& \text{\(P\)} \& \text{\(Q\)} \\ \hline \text{\(K\)} \& \text{\(L\)} \& \text{\(S\)} \\ \hline \text{\(\bar{Q} \)} \& \text{\(R\)} \\ \hline \end{tabular}$	<table><tr><td>L</td><td>P</td><td>Q</td></tr><tr><td>K</td><td>L</td><td>S</td></tr><tr><td>S</td><td>Q</td><td>R</td></tr></table>	L	P	Q	K	L	S	S	Q	R																												
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cardona-qft-methods_ TAB_007	286	0.474	$\begin{tabular}{t}{ l l } \hline \text{\(L\)} \& \text{\(p p p p\)} \\ \hline k \text{\(k k\)} \& \text{\(\bar{L} \)} \\ \hline \end{tabular}$	<table><tr><td>L</td><td>$pppp$</td></tr><tr><td>$k k k k$</td><td>L</td></tr></table>	L	$pppp$	$k k k k$	L	<table><tr><td>$k k$</td><td>$\bar{L}$</td></tr><tr><td>$k k$</td><td></td></tr></table> erator one has t that there are ey prediction o,	$k k$	$\bar{L}$	$k k$																													
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cardona-qft-methods_ TAB_008	287	■ 0.514	$\begin{tabular}[t]{t l l}\hline \mathfrak{s} \mathfrak{u}(2 \mid 2) & \text{create } (8_B + 8_F) \text{ magnons} \\ \hline \text{destroy } 8 + 8 \text{ magnons} & \overline{\mathfrak{s} \mathfrak{u}(2 \mid 2)} \end{tabular}$	<table><tr><td>$\mathfrak{su}(2 \mid 2)$</td><td>create $8_B + 8_F$ magnons</td></tr><tr><td>destroy 8 + 8 magnons</td><td>$\mathfrak{su}(2 \mid 2)$</td></tr></table>	$\mathfrak{su}(2 \mid 2)$	create $8_B + 8_F$ magnons	destroy 8 + 8 magnons	$\mathfrak{su}(2 \mid 2)$	e organised in a 8×8 matri: <table><tr><td>$\mathfrak{su}(2 2)$</td><td>create $8_B + 8_F$ magnons</td></tr><tr><td>destroy 8 + 8 magnons</td><td>$\overline{\mathfrak{su}(2 2)}$</td></tr></table>	$\mathfrak{su}(2 2)$	create $8_B + 8_F$ magnons	destroy 8 + 8 magnons	$\overline{\mathfrak{su}(2 2)}$
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